

Existence and Uniqueness Questions

Example 5: Determine the existence and uniqueness of the solutions to the system.

$$3x_1 - 6x_2 + 6x_4 + 4x_5 = -5$$

$$3x_1 - 7x_2 + 8x_3 - 5x_4 + 8x_5 = 9$$

$$3x_1 - 9x_2 + 12x_3 - 9x_4 + 6x_5 = 15$$

Solution: The augmented matrix of this system was row reduced in Example 3 to

$$\left[\begin{array}{ccccc|c} 3 & -9 & 12 & -9 & 6 & 15 \\ 0 & 2 & -4 & 4 & 2 & -6 \\ 0 & 0 & 0 & 0 & 1 & 4 \end{array} \right] \quad (8)$$

The basic variables are x_1, x_2 and x_5 ; the free variables are x_3 and x_4 . There is no equation, such as $0=1$ that would indicate an inconsistent system, so we could use back-substitution to find a solution. But the **existence** of a solution is already clear in (8). Also, the solution is **not unique** because there are free variables. Each different choice of x_3 and x_4 determines a different solution. Thus the system has infinitely many solutions. $\square$

When a system is in echelon form and contains no equation of the form $0=b$, with b nonzero, every nonzero equation contains a basic variable with a nonzero coefficient. Either the basic variables are completely determined (with no free variables) or at least one of the basic variables may be expressed in terms of one or more free variables. In the former case, there is a unique solution; in the latter case, there are infinitely many solutions (one for each choice of values for the free variables).

These remarks justify the following theorem.

Theorem 2: A linear system is consistent if and only if the rightmost column of the augmented matrix is not a pivot column—that is, if and only if an echelon form of the augmented matrix has no row of the form $[0 \dots 0 \ b]$ with b nonzero.